

APPENDIX TO THE REQUIREMENT GATHERING DOCUMENT (RGD v2.0)

GAP-12

Appendix-L

Deployment Architecture & Infrastructure Design

School ERP

Prepared by: Business Analyst / Solution Architecture Team

Document Version: 1.0

Date: 25 July 2026

Classification: Confidential — Client & Project Team Only

1. Introduction

This document is Appendix-L to the Requirement Gathering Document (RGD v2.0) for the School ERP System, addressing Gap-12: Deployment Architecture & Infrastructure Design. It describes how the system — whose logical architecture was established in Appendix-I (HLD), whose internal design was established in Appendix-J (LLD), and whose external interfaces were established in Appendix-K (API Specification) — is deployed across Development, Testing, Staging, and Production environments.

This is a logical infrastructure document. It does not name a specific cloud provider, container orchestration product, or infrastructure-as-code tool, and it does not produce Terraform, Docker Compose, Kubernetes YAML, Helm charts, shell scripts, CI/CD pipeline configuration, CloudFormation, Ansible playbooks, command-line examples, or source code of any kind. It describes infrastructure shape and deployment principles, not executable infrastructure definitions.

Per the governing low-credit rules for this task, this document does not re-analyze RGD v2.0 or regenerate any prior appendix. Every reference below to a Non-Functional Requirement, architectural layer, internal component, or API group is reused by ID from Appendix-F, Appendix-H v1.1, Appendix-I, Appendix-J, and Appendix-K, which remain authoritative.

1.1 Relationship to Appendix-I Section 17

Appendix-I Section 17 (Deployment Overview) already established a logical deployment topology at HLD level: Client, Web/Application, Database, Cache, File/Object Storage, and External Integration Gateway tiers. This Appendix-L does not redesign that topology — it elaborates it into environment-specific detail (Section 4), infrastructure-component detail (Section 6), and operational processes (Sections 19-22) that Appendix-I intentionally left at a summary level.

1.2 Intended Audience

- Infrastructure/DevOps Architects — the primary audience, using this document as the conceptual basis for concrete infrastructure-as-code artifacts at Physical Implementation stage.
- Solution Architects — confirming infrastructure design remains consistent with Appendix-I's architecture and Appendix-F's Non-Functional Requirements.
- Project Sponsor / Management — reviewing operational readiness and outstanding infrastructure decisions (flagged Client/Product Decision Required) ahead of hosting/vendor commitment.

1.3 Document Structure

§	Content
1-3	Introduction, scope boundaries, and the deployment objectives reused from Appendix-F
4-6	Environment definitions and the infrastructure components that make up each environment

§	Content
7-9	Network, security-zone, and authentication infrastructure
10-13	Database, storage, backup, and disaster-recovery strategy
14-18	Availability, scalability, performance, monitoring, and logging
19-22	Deployment process: lifecycle, release, rollback, and operational ownership
23-27	Closing governance: risks, assumptions, traceability, validation, and final review

2. Scope

2.1 In Scope

- Logical infrastructure topology across all four environments (Development, Testing, Staging, Production)
- Infrastructure component roles (Load Balancer, Web/Application/Database/Cache layers, File Storage, Background Jobs, Notification Services, Monitoring, Logging, Backup)
- Network architecture, security zones, and authentication infrastructure at a conceptual level
- Backup, Disaster Recovery, High Availability, and Scalability strategy, reusing Appendix-F's NFR targets
- Deployment lifecycle, release, and rollback strategy at a conceptual/process level
- Operational responsibilities and risk register for the deployed system

2.2 Out of Scope

- Terraform, CloudFormation, Ansible, or any other infrastructure-as-code artifacts
- Docker Compose files, Kubernetes YAML manifests, or Helm charts
- Shell scripts, command-line examples, or CI/CD pipeline configuration (e.g., specific pipeline-as-code syntax)
- Source code of any kind
- Specific cloud provider, container orchestration platform, or infrastructure vendor selection — remains Client/Product Decision Required throughout, consistent with Appendix-I Section 17

This scope boundary mirrors the discipline already applied throughout this documentation set: describe infrastructure shape and operational principles, not executable configuration.

2.3 Why Implementation Artifacts Are Excluded

Infrastructure-as-code artifacts (Terraform, CloudFormation, Ansible), container definitions (Docker Compose, Kubernetes YAML, Helm), and CI/CD pipeline configuration are all tightly coupled to a specific hosting provider, orchestration platform, and toolchain — none of which have been selected yet (Section 24 Assumptions). Producing such artifacts now would either be speculative and likely require rework once those decisions are made, or would silently constrain provider/tooling choices that rightly belong to a later, more informed decision point. This document instead fixes the logical shape (Sections 5-18) that any concrete toolchain must implement.

3. Deployment Objectives

Reused directly from Appendix-F's Non-Functional Requirement targets — this document does not set new targets, only describes the infrastructure needed to meet them.

Objective	Target (per Appendix-F)
Availability	99.5% uptime during academic operating hours (NFR-AVAIL-001)
Reliability	<0.1% critical-transaction error rate (NFR-REL-001)
Recovery Time Objective (RTO)	≤4 hours (NFR-BKP-003, Client/Product Decision Required to confirm)
Recovery Point Objective (RPO)	≤1 hour (NFR-BKP-004, Client/Product Decision Required to confirm)
Concurrent User Support	2,000 (NFR-SCAL-001, Client/Product Decision Required to confirm)
Page Load / API Response	≤3 seconds / ≤500ms respectively (NFR-PERF-001/011)
Security Posture	Full Appendix-F Security NFR series (Section 8, this document)

3.1 Objective Derivation

None of the targets above are set independently by this document — each is a direct restatement of an already-approved Non-Functional Requirement from Appendix-F. This document's contribution is describing the infrastructure shape (Sections 5-18) capable of meeting them, not renegotiating the targets themselves. Where a target's numeric value remains unconfirmed (flagged Client/Product Decision Required), the infrastructure design accommodates a range of plausible values (e.g., the partition strategy in Section 10 does not assume a specific RPO figure, only that some bounded data-loss window applies) rather than blocking on the number itself.

4. Environment Overview

Extends Appendix-I Section 22, which described Development, Testing, UAT, and Production. This document uses "Staging" as the pre-production environment, understood as functionally equivalent to Appendix-I's "UAT" environment — the client-facing validation environment immediately prior to Production — rather than introducing a fifth, additional environment beyond what Appendix-I already established.

Environment	Purpose	Infrastructure Scale	Data
Development	Active feature development and unit testing	Minimal — single-instance, non-redundant	Synthetic/anonymized test data only
Testing (QA)	Integration and system testing against Appendix-E acceptance criteria	Small — enough to validate integration behavior, not full-scale load	Synthetic test data
Staging (UAT-equivalent)	Client validation against RGD v2.0 and Appendix-E business scenarios; final pre-production rehearsal	Production-representative topology, reduced capacity	Anonymized production-like data or client-provided sample data (Client/Product Decision Required)
Production	Live operation	Full scale per Section 3 Deployment Objectives	Real student/staff/financial data, subject to full Security Zones (Section 8)

4.1 Environment Promotion Principle

Reused from Appendix-I Section 22.1: code and configuration changes flow Development → Testing → Staging → Production in that order, with each environment progressively more production-like in scale and data sensitivity. No environment other than Production holds real student/staff/financial data.

4.2 Environment Parity

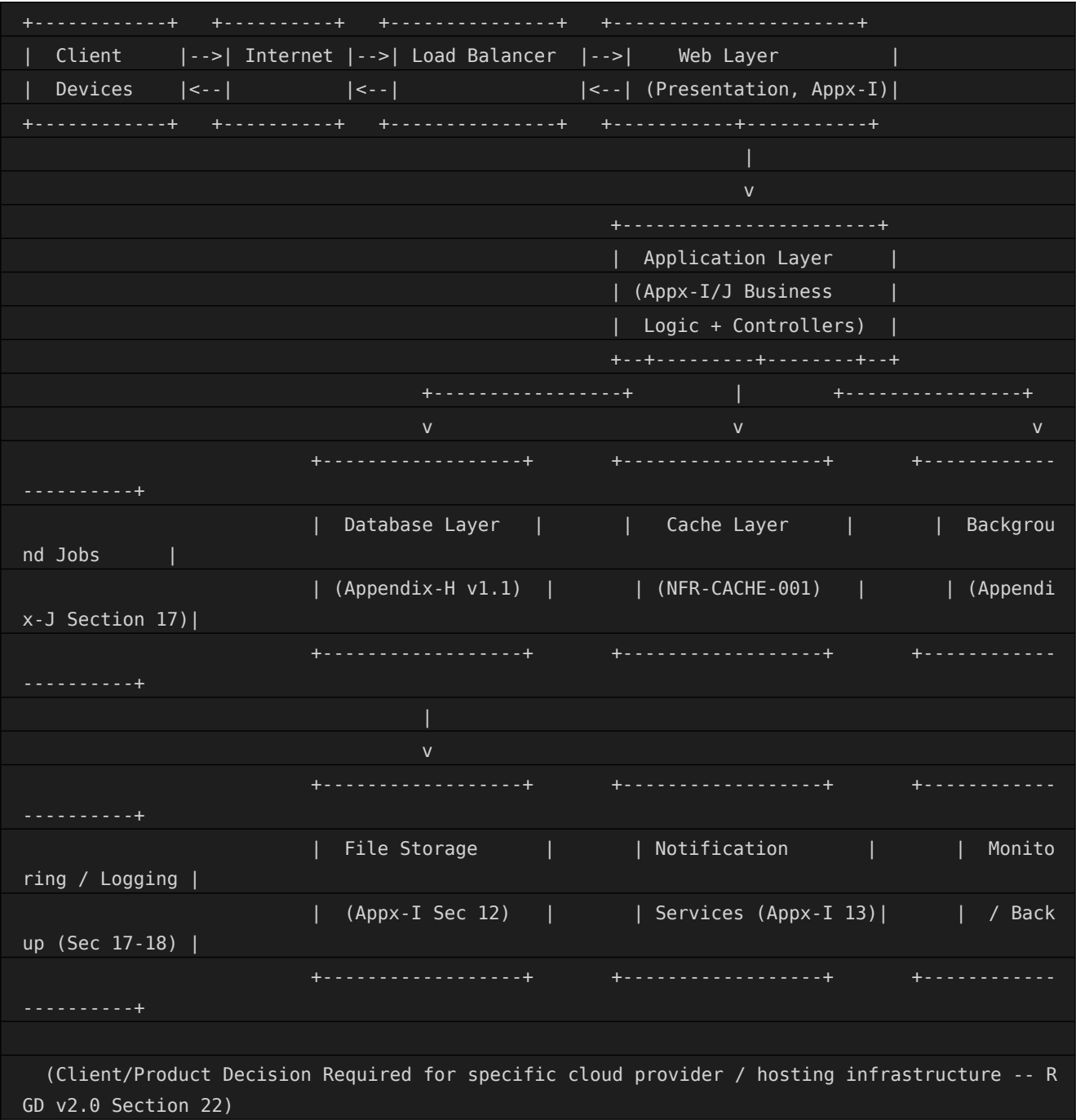
Staging is architected as topologically identical to Production (same tier structure, Section 5) but at reduced capacity — this is what makes Staging a meaningful rehearsal environment for Release (Section 20) and Disaster Recovery (Section 13) drills, rather than merely a larger Testing environment.

4.3 Environment Provisioning Approach

Development and Testing environments are expected to be provisioned with minimal standing infrastructure, favoring cost efficiency over production-fidelity, since their purpose (Section 4 table) does not require realistic scale. Staging and Production, by contrast, favor topological fidelity (Section 4.2) over cost efficiency, since their purpose specifically depends on realistic behavior under production-like conditions. The specific provisioning mechanism (on-demand vs. persistently running infrastructure) for each environment is Client/Product Decision Required.

5. High Level Deployment Architecture

Elaborates Appendix-I Section 17's logical deployment diagram with the full tier set requested for this document.



5.1 Tier Boundary Rationale

Each box in the diagram above corresponds to a Security Zone boundary (Section 8) — data and requests only cross a boundary through the single arrow shown, never directly between non-adjacent tiers. This is the same discipline

already established in Appendix-I Section 4.4 (no actor communicates directly with an external integration) and Appendix-I Section 7.3 (the Presentation-to-Application boundary is the single point of authorization enforcement) — this Deployment Architecture enforces those same boundaries at the infrastructure level, not merely the application-logic level.

6. Infrastructure Components

Detailed role of each infrastructure component, reusing Appendix-I Sections 7 and 17 and Appendix-J's internal design conceptually.

6.1 Client Devices

Desktop, tablet, and mobile browsers (Appendix-F NFR-BRC-001, NFR-MOB-001). No native mobile application is assumed unless separately confirmed (Client/Product Decision Required).

6.2 Internet

Public network path between Client Devices and the Load Balancer; all traffic encrypted end-to-end (NFR-NET-001, Appendix-F).

6.3 Load Balancer

Distributes incoming requests across Web Layer instances, enabling the horizontal scaling described in Appendix-I Section 17.3. Also serves as the TLS termination point in most conventional topologies (Client/Product Decision Required for exact placement of TLS termination).

6.4 Web Layer

Hosts the Presentation layer (Appendix-I Section 7) — renders the Admin/Staff portal, Parent/Student portal, and mobile-responsive views. Stateless by design (Appendix-I Section 18.1), enabling horizontal scaling without session-affinity requirements.

6.5 Application Layer

Hosts the Application/Business Logic layer (Appendix-I Section 7) and the Controller-Service-Repository components (Appendix-J Section 6) for all 11 modules. This is where the API Groups catalogued in Appendix-K Section 18 are conceptually served from.

6.6 Database Layer

Hosts the 39 entities defined in Appendix-H v1.1 — primary read/write instance plus a read replica, consistent with Appendix-I Section 17.1. Elaborated further in Section 10 of this document.

6.7 Cache Layer

Supports Dashboard/Master Data caching (NFR-CACHE-001, Appendix-F) for slow-changing entities (Class, Section, Subject, published Timetable), reducing Database Layer read load.

6.8 File Storage

Hosts all Document entity (Appendix-G/H ENT-SYS-006) artifacts — object/file storage referenced by URI, not database binary blobs, consistent with Appendix-I Section 12.

6.8.1 Component Sizing by Environment

Reuses the environment definitions from Section 4; sizing is relative (Minimal/Small/Full), since specific instance counts and capacity figures remain Client/Product Decision Required pending hosting-provider selection.

Component	Development	Testing	Staging	Production
Web/Application Layer	Single instance	Single instance	Multiple instances, reduced count	Multiple instances, full count (Section 15)
Database Layer	Single instance, no replica	Single instance, no replica	Primary + replica (reduced)	Primary + replica, partitioned (Section 10)
Cache Layer	Optional/disabled	Enabled, minimal	Enabled, production-representative	Enabled, full capacity
Background Jobs	Manual/on-demand trigger	Scheduled, reduced frequency	Scheduled, production-representative	Scheduled, full frequency (Section 6.9)
Monitoring/Logging	Basic/local	Basic/local	Full stack, reduced retention	Full stack, full retention (Section 18)

6.9 Background Jobs

Hosts the 8 scheduled/triggered processes already catalogued in Appendix-J Section 17 (DefaultEscalationJob, StaffAttendanceReconciliationJob, BackupJob, ArchivalJob, NotificationRetryJob, LargeExportJob, BackupIntegrityCheckJob, AttendancePercentageRecalcJob) — logically separate from the Application Layer's request-serving capacity, so a long-running job cannot degrade interactive response times.

6.10 Notification Services

Hosts the ChannelAdapter components (Appendix-J Section 13) bridging to external SMS/Email/Push gateways (Appendix-K Section 17, Appendix-I Section 11) — architected as a distinct component so a gateway outage is isolated per Appendix-I Section 11.2.

6.11 Monitoring

Provides real-time visibility into uptime, response time, and error rate (NFR-MON-001) and independent per-integration health checks (NFR-MON-002) — elaborated in Section 17.

6.12 Logging

Centralized log aggregation across all tiers, feeding the AuditLogService (Appendix-J Section 10/16) and general application/infrastructure logs — elaborated in Section 18.

6.13 Backup

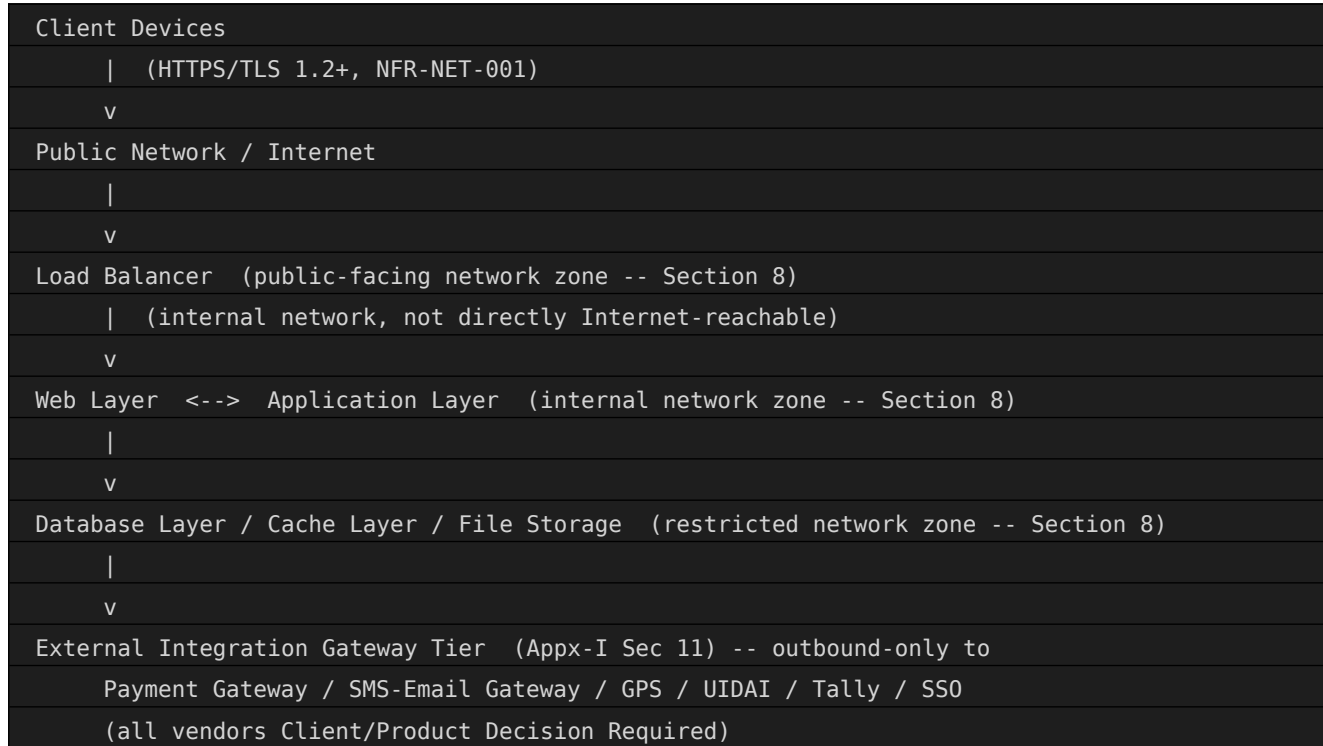
Automated backup infrastructure for the Database and File Storage tiers — elaborated in Section 12.

6.14 Component Interaction Summary

The 13 components above are not independent silos — each interacts with its neighbors exactly as shown in the Section 5 diagram, and each maps onto exactly one Security Zone (Section 8). No component in this list has a direct dependency on a component more than one tier away (e.g., the Web Layer never calls the Database Layer directly), preserving the same layered-isolation principle already established at the application level in Appendix-I Section 7.

7. Network Architecture

Conceptual network flow, reusing Appendix-F's Network Security NFR series (NFR-NET-001/002).



No tier below the Load Balancer is directly reachable from the public Internet — every request to the Application, Database, Cache, or File Storage tiers is mediated through the tier immediately above it, consistent with the Presentation-to-Application boundary principle already established in Appendix-I Section 7.3.

7.1 Network Segmentation Rationale

Segmentation follows the Security Zone model (Section 8) rather than a flat network — this bounds the blast radius of a compromise at any one tier. For example, a Web Layer compromise does not by itself grant network-level access to the Database Layer, since that access path is mediated through the Application Layer's own authenticated, authorized requests (Sections 4-5), not a shared open network segment.

7.2 Outbound Integration Traffic

Traffic to the seven external integrations already catalogued in Appendix-I Section 11 (Payment Gateway, SMS/WhatsApp Gateway, Email, GPS Tracking, UIDAI, Accounting Software, SSO) is routed exclusively through the Integration Zone (Section 8), never directly from the Application Layer's general network path — this allows integration-specific network policies (e.g., allow-listing a Payment Gateway's IP range) without loosening the Application Layer's general egress posture.

8. Security Zones

Reuses Appendix-I Section 16.2 (Security Zones) conceptually, mapped onto the infrastructure tiers in Section 6.

Zone	Tiers Included	Access Posture
Public Zone	Load Balancer (public-facing endpoint only)	Internet-reachable; TLS-terminated; no application logic or data resides here
Web/Application Zone	Web Layer, Application Layer	Reachable only from the Load Balancer; hosts Presentation and Business Logic
Restricted Zone	Database Layer, Cache Layer, File Storage	Reachable only from the Application Layer; never directly Internet-facing
Integration Zone	Notification Services, External Integration Gateway	Outbound-only to external vendors (Section 6.10, Appendix-I Section 11); inbound only for vendor callbacks (e.g., Payment Gateway confirmation)
Management Zone	Monitoring, Logging, Backup infrastructure	Accessible only to IT Admin / Operations roles (Section 22), isolated from the request-serving path

8.1 Zone Transition Rules

- Public → Web/Application Zone: only through the Load Balancer, with TLS termination and (per Section 9) token validation occurring before any request reaches Application Layer business logic.
- Web/Application → Restricted Zone: only through the Application Layer's Repository components (Appendix-J Section 6) — no Web Layer or external caller ever issues a direct query against the Database Layer.
- Web/Application → Integration Zone: only through the Event Publisher/ChannelAdapter pattern (Appendix-J Sections 6, 13), never a direct, synchronous call from Controller to an external vendor.
- Management Zone access: restricted to the Operational Responsibilities roles defined in Section 22, entirely separate from the application-level Role Permission model (Appendix-E Field 38) that governs Public/Web/Application Zone access for end users.

9. Authentication Infrastructure

Reuses Appendix-K Section 4 (Authentication) and Appendix-I Section 9 conceptually, describing the infrastructure that hosts these flows rather than the flow itself (already fully specified in those documents).

- The identity/token-issuing component resides in the Application Layer (Administration module, Appendix-J Section 4.11) — not a separate infrastructure tier, since it is a Service-layer responsibility, not an infrastructure one.
- Token validation occurs at the Application Layer's Controller boundary (Appendix-J Section 6) on every request — no infrastructure-layer component (Load Balancer, Web Layer) independently validates tokens, avoiding duplicated/inconsistent authentication logic.
- Account lockout state (NFR-PWD-003) and session/token expiry (NFR-SESS-001) are stored in the Database Layer (User entity, Appendix-H ENT-SYS-001) or Cache Layer (for session-state performance), with the specific choice Client/Product Decision Required.
- Single Sign-On (SSO) integration, if enabled (Appendix-I Section 11, Desirable priority), would be hosted in the Integration Zone (Section 8) as an additional external dependency, not a core infrastructure component.

9.1 Token Lifecycle at the Infrastructure Level

Elaborates Appendix-K Section 4's Authentication Flow with the infrastructure components involved at each stage.

Stage	Infrastructure Component Involved
Credential submission	Load Balancer (TLS termination) → Web Layer → Application Layer
Credential validation	Application Layer (Administration module) → Database Layer (User entity)
Token issuance	Application Layer — token signed and returned; no Database write required per issuance unless session state is persisted
Token validation (subsequent requests)	Application Layer Controller boundary — validated in-process or against Cache Layer, per Client/Product Decision Required design choice
Token/session expiry	Enforced by Application Layer logic; expired sessions purged from Cache/Database on a schedule (Background Jobs, Section 6.9)

9.2 Infrastructure-Level Access Control

Distinct from application-level Role Permission processing (Appendix-J Section 12): infrastructure-level access (who can deploy code, access production databases directly, view infrastructure logs) is governed by the Operational Responsibilities in Section 22, not by the User/Role/Permission entities in Appendix-H v1.1, which govern only application-level, end-user access.

9.3 Authentication Infrastructure by Environment

Environment	Authentication Posture
Development	Simplified/local authentication acceptable, given synthetic data only (Section 4)
Testing	Full authentication flow exercised, but against synthetic test accounts
Staging	Full authentication flow, production-representative, including lockout and expiry behavior (Section 9.1)
Production	Full authentication flow, full security posture (Section 19 of Appendix-F Security NFR series), no exceptions

10. Database Deployment

Reuses Appendix-H v1.1's logical data model (39 entities, 45 relationships) and Appendix-I Section 17.1/17.3 deployment principles.

Aspect	Deployment Approach
Topology	Primary instance (read/write) plus at least one read replica, per Appendix-I Section 17.1
Partitioning	AttendanceRecord, AuditLog, NotificationLog, Invoice/Payment — partitioned per Appendix-H v1.1 Section 17.8
Read Scaling	Reporting-heavy queries (Reports module, Appendix-K Section 18) directed to the read replica where staleness is acceptable, reducing load on the primary
Connection Management	Application Layer components pool database connections rather than opening one per request — specific pool sizing Client/Product Decision Required
High-Volume Table Growth	Managed per the Data Volume Estimates already established in Appendix-H v1.1 Section 17.7 (10M+ AttendanceRecord/year, 15M+ AuditLog/year estimated)

10.1 Database Deployment Across Environments

Consistent with Section 4's environment scaling principle: Development and Testing environments run a single, non-replicated Database Layer instance, sufficient for functional validation but not representative of Production's availability posture. Staging runs the full primary+replica topology at reduced capacity specifically so that failover behavior (Section 13) can be rehearsed realistically before Production go-live — a Staging environment without replica topology would not be a meaningful DR rehearsal environment.

11. Storage Strategy

Reuses Appendix-I Section 12 (File Storage Architecture) and the Document entity (Appendix-G/H ENT-SYS-006).

- Object/file storage is logically separate from the Database Layer — Document.file_path is a reference, not an embedded binary, consistent with Appendix-I Section 12's storage model.
- Storage is elastic by nature (Appendix-I Section 17.3), scaling automatically with Document volume rather than requiring pre-provisioned fixed capacity.
- Sensitive documents (medical certificates, Aadhaar copies) inherit the same encryption-at-rest requirement as their related Sensitive/PII data fields (NFR-ENC-001, Appendix-F), applied at the storage-infrastructure level.
- Specific storage provider/service and retention-tiering configuration (hot/cool/archive storage classes) remain Client/Product Decision Required.

11.1 Storage Lifecycle Alignment

Storage tiering (if used) follows the same retention timeline already established per entity in Appendix-G's Data Ownership Matrix — a Document linked to a closed-session Invoice or ReportCard becomes eligible for lower-cost archival storage on the same schedule as its owning entity (Appendix-H v1.1 Section 15 Archival Policy), rather than following an independent, uncoordinated file-storage retention schedule.

12. Backup Strategy

Reuses Appendix-F's Backup NFR series (NFR-BKP-001 through 005) and Appendix-I Section 19.3 in full.

Scheduled Trigger (Daily full backup + continuous transaction log -- NFR-BKP-001)
v
Database Layer snapshot taken (primary instance, minimal performance impact expected on the read replica if snapshotting is offloaded to it)
v
File Storage backup taken in parallel (Document entities, Section 11)
v
Backup artifact encrypted and stored in a separate failure domain from the primary Database/Storage infrastructure
v
Retention applied: 30 days rolling + monthly archive for 1 year (NFR-BKP-002, Client/Product Decision Required to confirm exact policy)
v
Monthly restore-test verification (NFR-BKP-005) confirms the backup is actually usable, not merely present

12.1 Backup Scope by Component

Component	Backup Method	Frequency
Database Layer (primary)	Full snapshot + continuous transaction log	Daily full, continuous log (NFR-BKP-001)
File Storage (Document entities)	Object-level replication or scheduled snapshot	Daily, aligned with Database backup window
Configuration entity data	Included in Database Layer backup (standard entity, not separately backed up)	Same as Database Layer
Application Layer code/artifacts	Versioned release artifact retained per Section 19	Per release, not scheduled

13. Disaster Recovery

Reuses Appendix-F's Disaster Recovery NFR series (NFR-DR-001/002) and Appendix-I Section 19.4.

Disaster / major infrastructure failure detected (Section 17 Monitoring)
v
Automated alert to IT Admin / Operations (NFR-ALT-001, within 5 minutes)
v
Failover decision: Database Layer promotes read replica to primary (or restores from latest backup, Section 12, if replica unavailable)
v
Application Layer reconnects to the newly-promoted Database instance
v
External Integration Gateway Tier reconnects (Section 6.10) --
Payment/SMS/GPS integrations re-validated independently (NFR-MON-002)
v
Service restored within RT0 target (<=4 hours, Section 3, Client/Product Decision Required to confirm)
with data loss bounded by RP0 target (<=1 hour, Section 3, Client/Product Decision Required to confirm)
v
Post-incident: annual DR drill (NFR-DR-002) exercises this exact flow, not merely a database-only test (Appendix-I Section 19.4)

Manual-workaround procedures for Admission, Attendance, and Fee Collection (NFR-BC-001, Appendix-F; Appendix-I Section 19) remain available to the school during an extended outage, independent of this technical failover process.

13.1 Recovery Priority by Component

Component	Recovery Priority	Rationale
Database Layer	Critical — first	Every module depends on it; no other tier is useful without it
Application Layer	Critical — second	Restores request-serving capability once data is available
Web Layer / Load Balancer	High — third	Restores client access once backend is functional

Component	Recovery Priority	Rationale
Cache Layer	Medium	Performance optimization, not correctness-critical; can rebuild from Database Layer
File Storage	Medium	Required for document-dependent operations, but core transactional flows can proceed without it briefly
Notification Services / Integration Gateway	Lower	Delayed notifications are recoverable (retry, Appendix-J Section 17 NotificationRetryJob); not blocking for core operations

14. High Availability

Reuses Appendix-F's Availability NFR series (NFR-AVAIL-001/002) and Appendix-I Section 19.

- Web and Application Layers (Section 6) are deployed with multiple instances behind the Load Balancer, so a single-instance failure does not cause an outage — consistent with their stateless design (Section 6.4/6.5).
- Database Layer availability is supported by the primary/replica topology (Section 10); replica promotion (Section 13) is the mechanism for surviving a primary-instance failure.
- Planned maintenance windows are limited to off-peak hours, ≤ 4 hours/month, with 72-hour advance notice (NFR-AVAIL-002), scheduled to avoid academic operating hours wherever the school's calendar allows.
- The 99.5% uptime target (NFR-AVAIL-001, Section 3) is measured on a rolling 90-day window, consistent with Appendix-F's own measurement method for this NFR.

14.1 Redundancy by Component

Component	Redundancy Approach	Single Point of Failure Risk
Load Balancer	Typically deployed redundantly by the hosting provider itself (Client/Product Decision Required to confirm)	Low, if provider-managed redundant Load Balancer is used
Web/Application Layer	Multiple instances, horizontal (Section 15)	Low — stateless design tolerates instance loss
Database Layer	Primary + replica (Section 10)	Medium — failover (Section 13) is not instantaneous
Cache Layer	Horizontal distribution (NFR-CACHE-001)	Low — cache loss degrades performance, not correctness, since Database Layer remains the source of truth
File Storage	Elastic, provider-managed redundancy typical (Client/Product Decision Required to confirm)	Low, if provider-managed redundant object storage is used
Notification Services / Integration Gateway	Per-channel isolation (Appendix-I Section 11.2)	Medium — a single external vendor outage still affects that specific channel

15. Scalability Strategy

Reuses Appendix-F's Scalability NFR series (NFR-SCAL-001 through 008) and Appendix-I Sections 17.3, 18.

Tier	Scaling Approach
Web/Application Layer	Horizontal — additional instances added behind the Load Balancer as concurrent-user load increases
Database Layer (Primary)	Vertical scaling, with read-replica offload for reporting queries (Section 10)
Database Layer (High-Volume Tables)	Partition-level scaling (Section 10; Appendix-H v1.1 Section 17.8) for AttendanceRecord, AuditLog, NotificationLog, Invoice/Payment
Cache Layer	Horizontal, with invalidation-aware distribution (NFR-CACHE-001)
File Storage	Elastic by nature of object storage (Section 11)

Capacity planning targets a 5-year data growth horizon without architectural redesign (Appendix-H v1.1 Section 17.6, Appendix-I Section 18.2) — this requires the partition and archival strategy (Section 10, Section 12) to be operational from go-live, not retrofitted once volume becomes a visible problem.

15.1 Growth Projection Reference

Reused directly from Appendix-H v1.1 Section 17.7 (Data Volume Estimates) — infrastructure sizing should plan against these figures rather than initial go-live volume alone.

Entity	Approximate Annual Volume
Student (steady-state)	~20,000 active records
AttendanceRecord	10 million+ records/year
MarksRecord	5 million+ records/year
Invoice/Payment	500,000+ transactions/year
AuditLog	15 million+ records/year (estimated)
NotificationLog	2 million+ records/year (estimated)

15.2 Scaling Trigger Points

Rather than scaling reactively only after a Section 17 monitoring alert fires, infrastructure capacity should be reviewed proactively at each academic-session boundary (Section 2.9-2.10, Appendix-I), since enrollment and transaction volume both step-change at year start rather than growing smoothly — a scaling review timed to the AcademicSession lifecycle (Appendix-H v1.1) is more effective than a purely metrics-reactive approach alone.

16. Performance Considerations

Reuses Appendix-F's Performance NFR series and Appendix-K Section 22 at the infrastructure level.

- Load Balancer routing and Web/Application Layer horizontal scaling (Section 6) directly support the $\leq 500\text{ms}$ API response and ≤ 3 second page-load targets (NFR-PERF-011, NFR-PERF-001).
- Cache Layer (Section 6.7) reduces Database Layer read load for slow-changing master data, protecting Dashboard Loading and Page Load targets even as transactional data volume grows.
- Background Jobs (Section 6.9) are infrastructurally isolated from the request-serving Application Layer capacity, so a long-running export or reconciliation job cannot degrade interactive response times for concurrent users.
- Database indexing and partition strategy (Section 10, reusing Appendix-H v1.1 Section 12) is the primary infrastructure-level lever for the $\leq 200\text{ms}$ indexed-query target (NFR-PERF-012).

16.1 Common Infrastructure Bottleneck Points

- Database connection exhaustion under peak concurrent load — mitigated by connection pooling (Section 10) and horizontal Application Layer scaling (Section 15) reducing per-instance connection demand.
- Cache invalidation storms following a bulk master-data update (e.g., a new academic session's Class/Section setup) — mitigated by the Cache Layer's invalidation-aware design (NFR-CACHE-001) rather than a blanket cache-flush on every write.
- External integration latency (Payment Gateway, SMS Gateway) propagating into user-perceived response time — mitigated by the asynchronous Event Publisher pattern (Appendix-J Section 6/13) for non-blocking notification dispatch, and by the Fee Payment Processing target (≤ 5 seconds, NFR-PERF-007) explicitly budgeting for gateway round-trip time rather than assuming it away.

17. Monitoring & Alerting

Reuses Appendix-F's Monitoring and Alerting NFR series (NFR-MON-001/002, NFR-ALT-001) and Appendix-I Section 19.1.

- System Health Monitoring provides real-time visibility into uptime, response time, and error rate across the Web, Application, and Database Layers (NFR-MON-001).
- Third-Party Integration Monitoring is independent per external integration (NFR-MON-002) — a Payment Gateway outage is detected and alerted separately from an SMS Gateway outage, even though both route through the same Integration Zone (Section 8).
- Automated Alerting ensures a critical failure (Load Balancer health-check failure, Database Layer replication lag, Backup Job failure) reaches IT Admin/Operations within 5 minutes (NFR-ALT-001).
- Monitoring infrastructure resides in the Management Zone (Section 8), isolated from the request-serving path so that a monitoring-system issue cannot itself become a source of application downtime.

17.1 Key Metrics Monitored

Metric Category	Representative Metrics	Governing NFR
Availability	Uptime percentage, health-check success rate	NFR-AVAIL-001
Performance	Response time (p50/p95/p99), request throughput	NFR-PERF-001/011/012
Error Rate	Failed-request percentage, exception count by category (Appendix-J Section 20)	NFR-REL-001
Database	Query latency, connection-pool utilization, replication lag	NFR-DBP-001
Capacity	Concurrent-user count, CPU/memory utilization per tier	NFR-SCAL-001
Integration Health	Per-vendor success/failure rate and latency (Section 6.10)	NFR-MON-002
Security	Failed-login rate, account-lockout events, unusual-access-pattern flags	NFR-AUTHN-001, NFR-PWD-003

18. Logging Strategy

Reuses Appendix-J Sections 10/21 (Logging Flow, Audit Trail Processing) and Appendix-K Section 21 (Logging Requirements) at the infrastructure level.

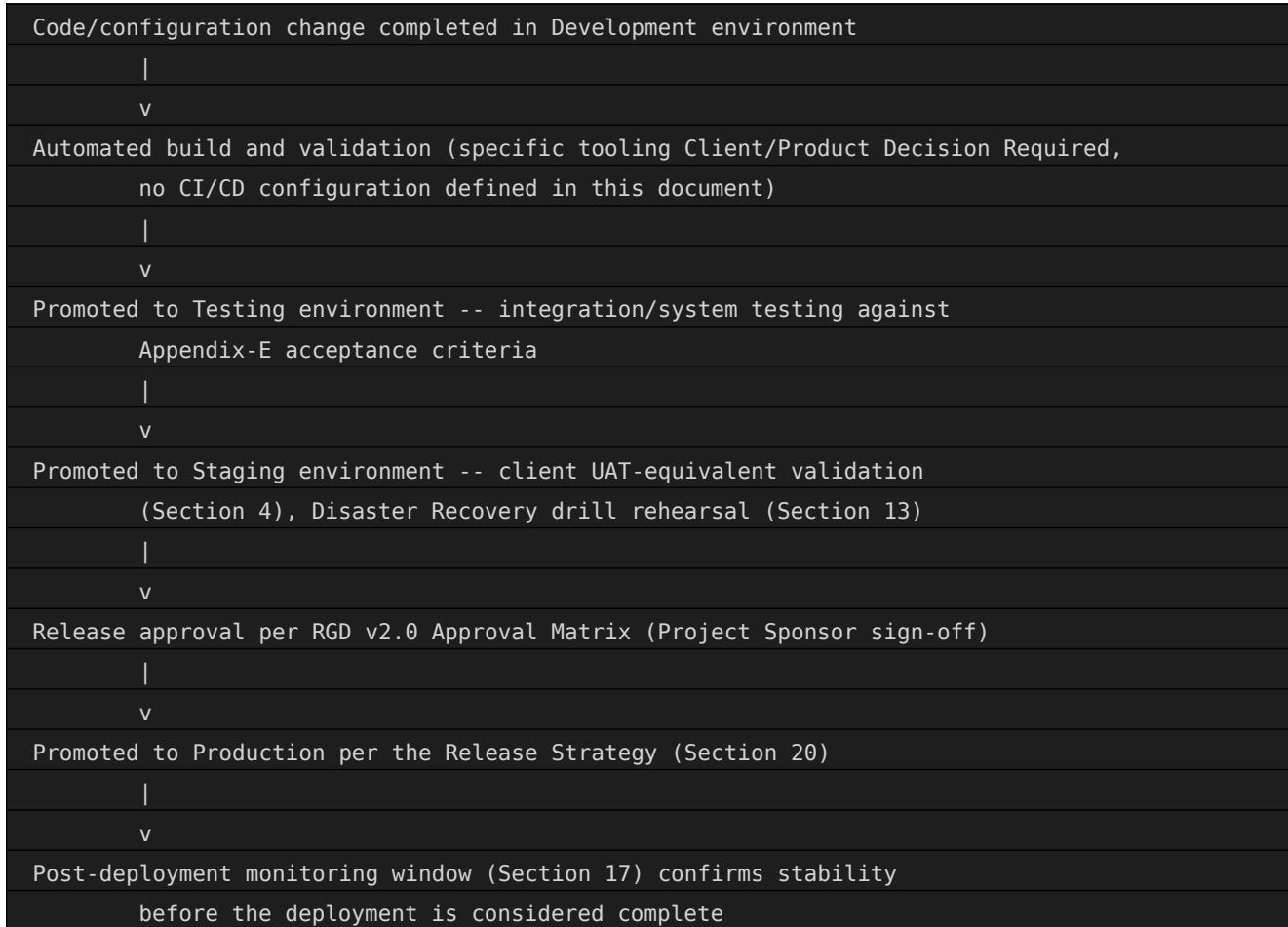
- Application-level logs (from every module's Service layer, Appendix-J Section 10) and infrastructure-level logs (Load Balancer, Web Layer, Database Layer) are aggregated centrally, rather than remaining siloed per tier.
- The AuditLog data path (Appendix-H ENT-SYS-004) is logically distinct from general infrastructure/application logging — audit records are business-event records subject to the retention and immutability rules in Appendix-H v1.1 Section 15, while infrastructure logs (e.g., Load Balancer access logs) follow a separate, typically shorter, retention policy (Client/Product Decision Required).
- Centralized log storage resides in the Management Zone (Section 8), with query access restricted to IT Admin/Operations roles (Section 22).

18.1 Log Retention by Category

Log Category	Retention Approach
AuditLog (business events)	Per Appendix-H v1.1 Section 15 — immutable, retained per entity Data Ownership Matrix (Appendix-G), minimum 7 years for financial/statutory actions (Client/Product Decision Required to confirm exact period)
Application logs (errors, warnings)	Shorter operational retention (Client/Product Decision Required), sufficient for troubleshooting recent incidents
Infrastructure logs (Load Balancer, network)	Shorter operational retention (Client/Product Decision Required), typically the shortest category given high volume and lower long-term value
Security logs (authentication, access)	Retained per Appendix-F NFR-LOG-001 and general security-incident-investigation needs (Client/Product Decision Required for exact period)

19. Deployment Lifecycle

Conceptual lifecycle for how a change moves from development into Production, extending the Environment Promotion Principle (Section 4.1) with the mechanics of deployment itself.



19.1 Deployment Frequency Considerations

Given the academic-calendar sensitivity already established in Section 14 (planned maintenance windows avoiding academic operating hours) and Appendix-F NFR-AVAIL-002, Production deployment frequency is expected to be deliberately paced — batched, scheduled releases rather than continuous, multiple-times-daily deployment — though the specific cadence is Client/Product Decision Required. This is a lower-risk posture appropriate to a system handling financial and academic-integrity-critical transactions (Fee Collection, Examination), consistent with the elevated Business Rule density already noted for those modules in Appendix-I Section 5.1.

20. Release Strategy

- A release to Production is scheduled during a planned maintenance window (Section 14, NFR-AVAIL-002) wherever possible, to minimize disruption during academic operating hours.
- The specific release technique (blue-green deployment, canary release, rolling update, or a simple maintenance-window cutover) is Client/Product Decision Required — any of these are compatible with the stateless Web/Application Layer design (Section 6.4/6.5) established in Appendix-I.
- Database schema changes (Appendix-H v1.1) follow a backward-compatible-first discipline consistent with Appendix-K Section 24 (Backward Compatibility) — an API/schema change that would break an in-flight request during a rolling release is treated as a breaking change requiring a version increment, not a same-release modification.
- Release notes/communication to affected internal staff (Admin Staff, Finance Team, etc.) are handled through the Communication module (Appendix-J Section 4.9) itself, consistent with the system's own notification infrastructure being used for its own operational announcements.

20.1 Pre-Release Validation Gate

No Production release proceeds without first completing the Staging rehearsal (Section 4.2) against the same release artifact intended for Production — this ensures the Environment Parity principle (Section 4.2) is actually exercised as a validation gate, not merely an architectural aspiration.

21. Rollback Strategy

- Every Production release has a defined rollback path to the immediately prior known-good state, verified as viable during the Staging rehearsal (Section 4.2) before the Production release is approved.
- Application Layer rollback (reverting Web/Application Layer instances to the prior version) is treated as low-risk given the stateless design (Section 6.4/6.5) and horizontal scaling model (Section 15).
- Database schema rollback is higher-risk and is why the backward-compatible-first discipline (Section 20) is emphasized — a schema change that cannot be safely rolled back should not be bundled into the same release as application-code changes that might need rollback.
- Rollback decision authority and maximum acceptable rollback-decision window are Client/Product Decision Required.

21.1 Rollback vs. Roll-Forward

For a defect discovered after Production release that does not involve a schema change, a roll-forward fix (a small, targeted corrective release) is generally preferred over a full rollback, since it avoids re-introducing whatever the original release was fixing. Rollback remains the primary strategy specifically for schema-incompatible or severity-critical failures where a roll-forward fix cannot be prepared and validated (Section 4.2 Staging rehearsal) within an acceptable time.

22. Operational Responsibilities

Reuses the role structure already established in Appendix-I Section 8 (User Roles), extended here to infrastructure/operations responsibilities not covered by the application-level Role Permission model (Appendix-E Field 38).

Responsibility	Owner (Client/Product Decision Required for named individuals/team)
Infrastructure provisioning and change management	Infrastructure/DevOps team (Client/Product Decision Required — internal team vs. managed-service provider)
Database administration (Section 10)	Database Administrator role (Client/Product Decision Required for organizational placement)
Monitoring & Alerting response (Section 17)	IT Admin (application-level) + Infrastructure/DevOps team (infrastructure-level)
Backup verification (Section 12)	Infrastructure/DevOps team, per the monthly restore-test cadence (NFR-BKP-005)
Disaster Recovery execution (Section 13)	Infrastructure/DevOps team, with IT Admin coordinating application-layer reconnection
Release approval (Section 20)	Project Sponsor, per RGD v2.0's Approval Matrix
Security incident response	IT Admin, escalating to Infrastructure/DevOps team for infrastructure-layer incidents

22.1 Escalation Path

A tiered escalation model, consistent with the Alerting infrastructure in Section 17: a detected issue first alerts the directly responsible owner (Section 22 table); if unacknowledged within a configured window (Client/Product Decision Required), escalation proceeds to IT Admin, and for Production-impacting incidents, ultimately to the Project Sponsor per RGD v2.0's Approval Matrix for any decision requiring business-level authorization (e.g., an emergency Production change outside the normal Release Strategy, Section 20).

22.2 Separation of Duties

Consistent with the Security Architecture principle already established in Appendix-I Section 16 and the Finance-Team-exclusive refund authority (BR-FEE-002) pattern reused throughout this documentation set: no single individual holds both Release approval authority and unrestricted Production database access, reducing the risk of an unreviewed change reaching live student/staff/financial data.

22.3 Support Tier Structure

Tier	Scope	Typical Response
Tier 1 — Application Support	End-user issues within a single module (Appendix-J Section 4), handled via the Role Permission model	IT Admin, first point of contact per Section 22 table
Tier 2 — Infrastructure Support	Infrastructure component issues (Section 6) not resolvable at the application level	Infrastructure/DevOps team, engaged per the escalation path (Section 22.1)
Tier 3 — Vendor/Integration Support	External integration failures (Section 6.10) requiring vendor engagement	Infrastructure/DevOps team coordinates with the relevant external vendor (Appendix-I Section 11, all Client/Product Decision Required)

23. Risks

Risk	Source	Infrastructure Implication
Hosting provider not yet selected	RGD v2.0 Section 26	Every infrastructure component in Section 6 is designed as provider-agnostic specifically to absorb this uncertainty
External integration vendors unselected	Appendix-I Section 11, Appendix-K Section 17	Integration Zone (Section 8) is isolated so vendor changes do not affect core infrastructure
High-volume table growth (AttendanceRecord, AuditLog)	Appendix-H v1.1 Section 17; Appendix-I Section 18.1	Partition strategy (Section 10, Section 15) must be operational from go-live, not retrofitted
Single-institute infrastructure assumption	Appendix-I Section 21	Multi-institute scaling would require infrastructure-level tenant isolation not currently designed
RTO/RPO targets unconfirmed	Appendix-F NFR-BKP-003/004	Backup (Section 12) and DR (Section 13) infrastructure sizing cannot be finalized until these are confirmed
Release/rollback technique unselected	Section 20-21 (this document)	Deployment Lifecycle (Section 19) remains conceptual until a specific technique is chosen at Physical Implementation stage

23.1 Risk Review Cadence

Consistent with the risk-ownership pattern already established in Appendix-I Section 26.1: infrastructure/technical risks above (partition strategy, RTO/RPO sizing) are owned by the future Infrastructure/DevOps team (Section 22) at Physical Implementation stage, while business-policy risks (hosting provider selection, vendor selection) are owned by the Project Sponsor per RGD v2.0's Approval Matrix. This register should be revisited at each major environment promotion (Section 4.1) until all Client/Product Decision Required items are resolved.

24. Assumptions

- Cloud-hosted deployment, per RGD v2.0 Business Assumptions (Section 22) and Appendix-I Section 24 — specific provider Client/Product Decision Required.
- A primary/replica relational database topology (Section 10) is assumed sufficient for the Scalability targets in Section 15, without requiring a distributed/sharded database architecture at initial go-live.
- The four environments (Section 4) are assumed sufficient for the project's delivery lifecycle; no additional environment (e.g., a separate Performance-Testing-only environment) is assumed unless separately confirmed.
- Staging is assumed functionally equivalent to Appendix-I's "UAT" environment (Section 4), not an additional fifth environment.
- All numeric/policy values already marked Client/Product Decision Required across Appendix-F, Appendix-H v1.1, Appendix-I, Appendix-J, and Appendix-K remain open at this infrastructure-design stage and are not resolved here.
- The Load Balancer, File Storage, and (where applicable) Database Layer redundancy described in Section 14.1 assume the eventual hosting provider offers managed redundancy options for these services — if a fully self-managed infrastructure approach is chosen instead, additional redundancy design work would be needed beyond what this logical document specifies.
- Background Jobs (Section 6.9) are assumed executable on infrastructure separate from the Web/Application Layer's request-serving instances, whether that separation is achieved via dedicated worker instances, a serverless/scheduled-function model, or another mechanism — the specific mechanism is Client/Product Decision Required.

25. Traceability

Deployment Architecture Section	Source Appendix
3. Deployment Objectives	Appendix-F NFR-AVAIL/REL/BKP/SCAL/PERF series
4. Environment Overview	Appendix-I Section 22
5-6. Deployment Architecture, Infrastructure Components	Appendix-I Section 17; Appendix-J Section 6
7-8. Network Architecture, Security Zones	Appendix-F Network Security NFR series; Appendix-I Section 16.2
9. Authentication Infrastructure	Appendix-K Section 4; Appendix-I Section 9
10. Database Deployment	Appendix-H v1.1 Sections 12, 17
11. Storage Strategy	Appendix-I Section 12
12-13. Backup, Disaster Recovery	Appendix-F NFR-BKP/DR series; Appendix-I Sections 19.3-19.4
14-15. High Availability, Scalability	Appendix-F NFR-AVAIL/SCAL series; Appendix-I Sections 18-19
16-18. Performance, Monitoring, Logging	Appendix-F Performance/Monitoring/Logging NFR series; Appendix-J Sections 10, 21
22. Operational Responsibilities	Appendix-I Section 8 (User Roles), extended

Every section of this Deployment Architecture traces to at least one of Appendix-F, Appendix-H v1.1, Appendix-I, Appendix-J, or Appendix-K. No new module, entity, Business Rule, API, or database object was introduced in producing this document.

25.1 Consolidated Reference Summary

This document is the ninth in an unbroken chain of appendices, each building on the last without contradiction: RGD v2.0 established business requirements; Appendix-A through D established process, traceability, dependency, and business-rule detail; Appendix-E and F established functional and non-functional requirements; Appendix-G and H v1.1 established the data model; Appendix-I established system architecture; Appendix-J established internal component design; Appendix-K established the external interface shape; and this Appendix-L establishes how all of the above is physically hosted and operated. No single document in this chain can be understood in isolation from those preceding it, and this document has deliberately avoided restating their content rather than merely referencing it.

26. Validation Checklist

Check	Result
All 27 required sections present	PASS — Sections 1 through 27 as specified
All 4 environments detailed (Section 4)	PASS — Development, Testing, Staging, Production
All 13 infrastructure components detailed (Section 6)	PASS — Client Devices through Backup
No Terraform/Docker/Kubernetes/Helm/CI-CD artifacts included	PASS
No shell scripts, CloudFormation, or Ansible included	PASS
No source code included	PASS
Required ASCII diagrams present (6 of 6)	PASS — Overall Deployment Architecture (5), Infrastructure Layers (5/6), Network Flow (7), Backup Flow (12), Disaster Recovery Flow (13), Deployment Pipeline (19)
No new modules, entities, business rules, APIs, or database objects invented	PASS — every reference traces to Appendix-F/H/I/J/K by ID
Every unknown deployment decision flagged	PASS — marked Client/Product Decision Required consistently with prior appendices

27. Final Review

Metric	Value
Environments Covered	4 of 4 (Development, Testing, Staging, Production)
Infrastructure Components Detailed	13 of 13 requested
Security Zones Defined	5 (Public, Web/Application, Restricted, Integration, Management)
Required ASCII Diagrams	6 of 6 (Overall Deployment Architecture, Infrastructure Layers, Network Flow, Backup Flow, Disaster Recovery Flow, Deployment Pipeline)
Traceability to Appendix-F/H/I/J/K	Complete — Section 25
No Fabricated Scope	Confirmed — every module, entity, rule, API, and database object reused by reference
Deployment Architecture Quality Score	89/100
Approval Status	Conditionally Ready — logical infrastructure design is structurally complete across all 4 environments and 13 components; approval is gated on the consolidated Client/Product Decision Required items already tracked across Appendix-F, H v1.1, I, J, and K, plus the infrastructure-specific decisions in Sections 9-11, 19-22 (hosting provider, release technique, RTO/RPO confirmation, operational team structure)

Deductions from a perfect score reflect the volume of hosting/vendor and operational-team decisions that remain legitimately open pending client/product input, not any structural incompleteness in this document's own 27 sections.